EART27201 Sedimentary Rocks and Fossils



Macrofossils: Brachiopods and Molluscs

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Macrofossils: Brachiopods

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Introduction and biology

What is a Brachiopod?

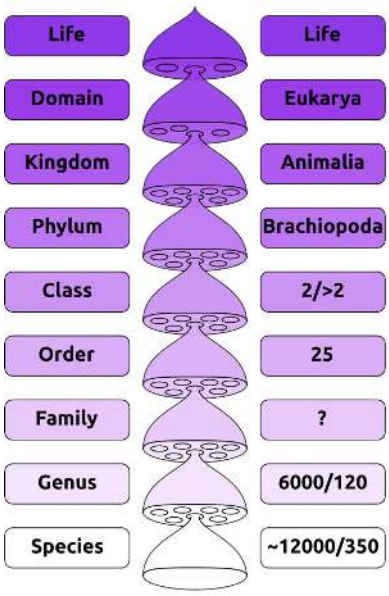


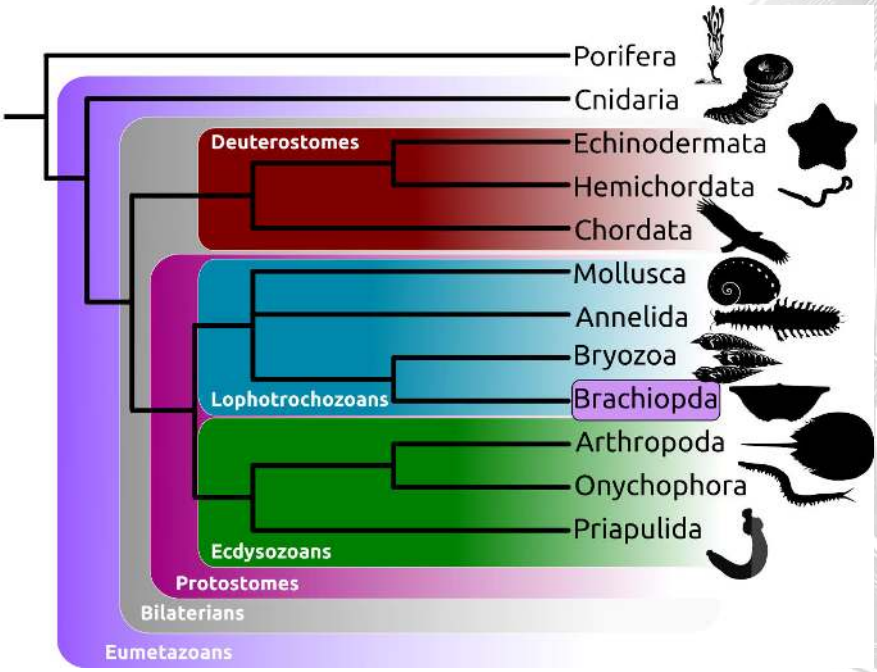


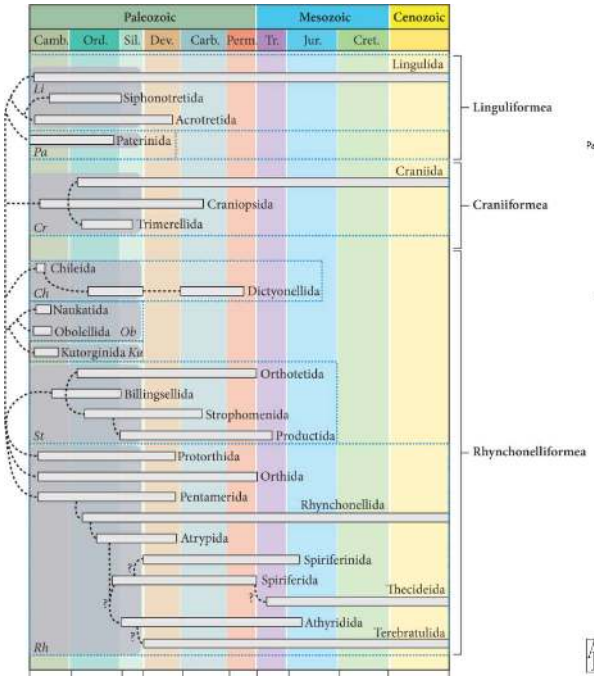


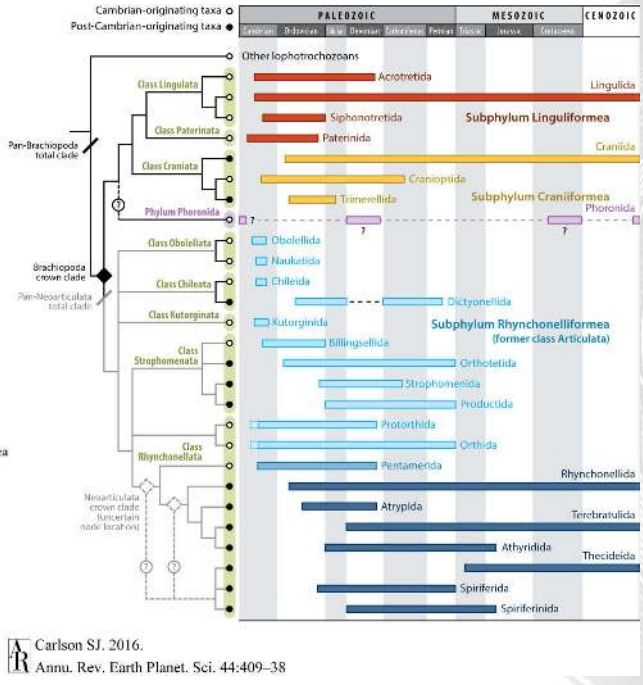
Images by: James St. John (left, middle); Mark A. Wilson (right).

Where do they sit on the tree of life?



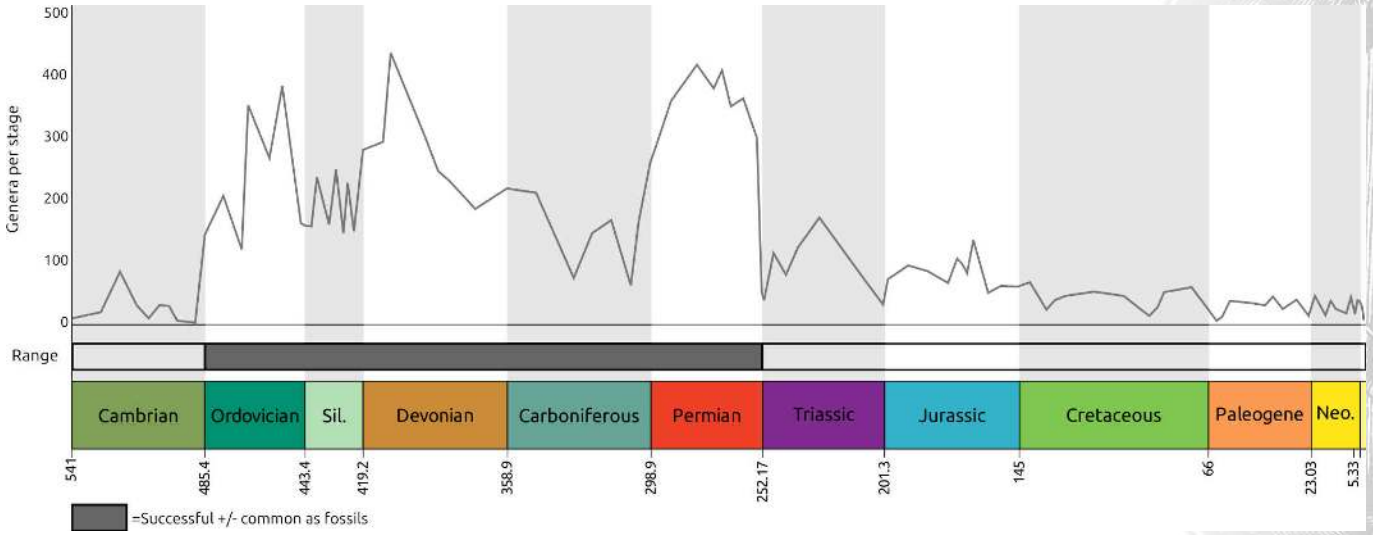






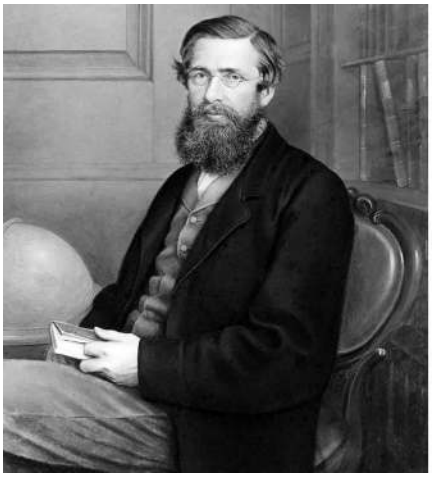
Left: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.
Right: Carlson, S.J., 2016. The evolution of Brachiopoda. Annual Review of Earth and Planetary Sciences, 44, pp.409-438.

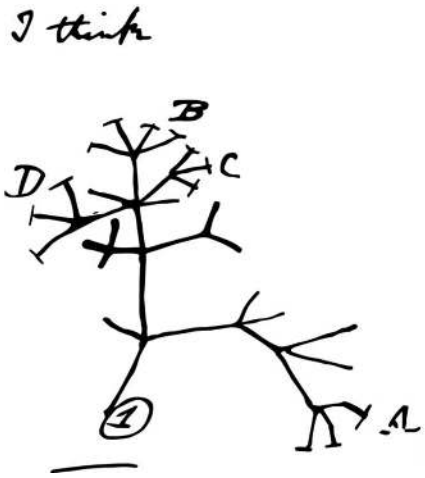
When were brachiopods successful?



Fossil data from PBDB.

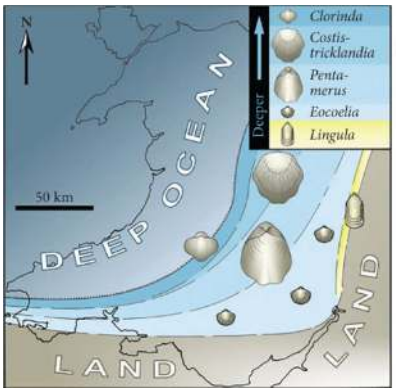
Why do they matter?

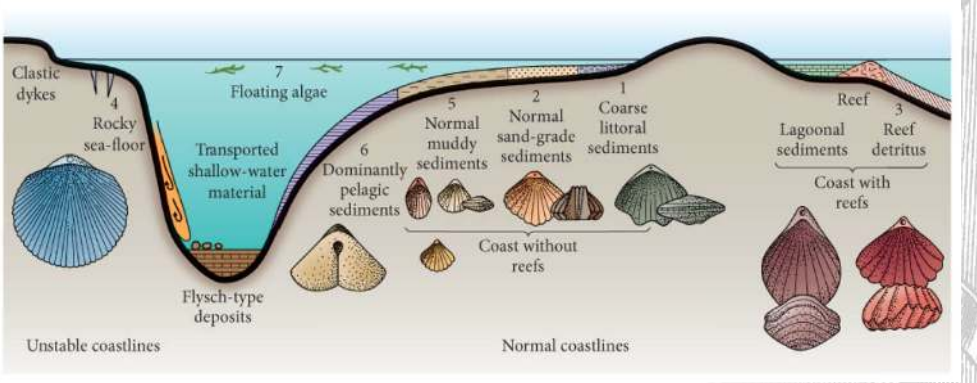






Alfred Russel Wallace and Charles Darwin





Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.

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Macrofossils: Brachiopods

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Morphology / Anatomy

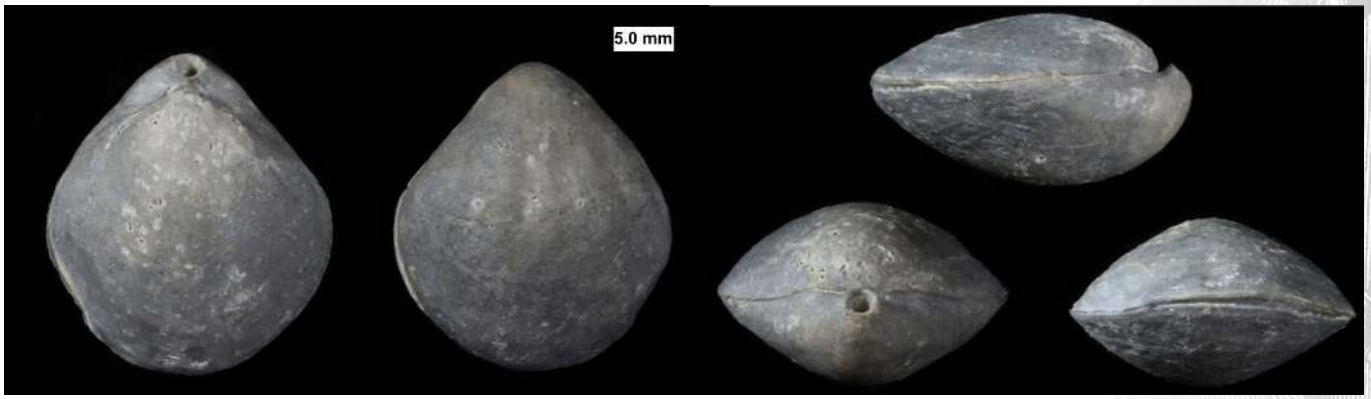
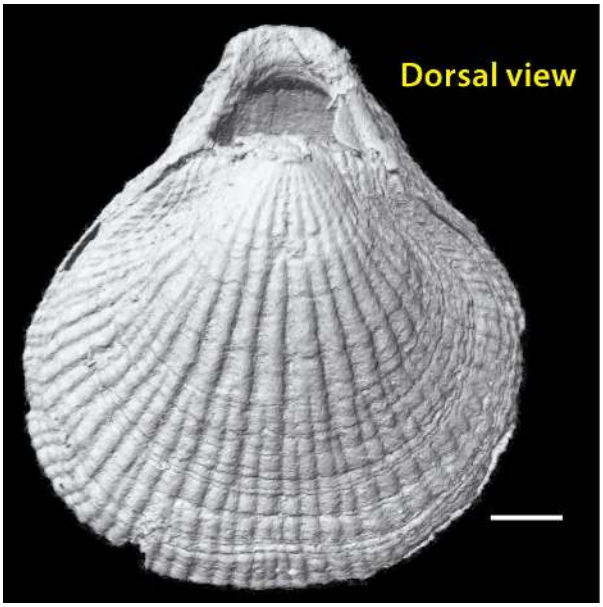


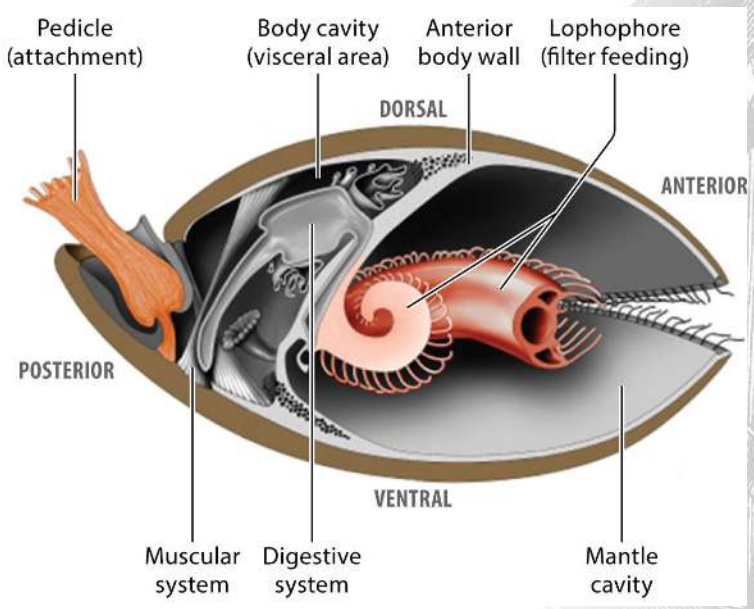
Image by Kenneth C. Gass

Ventral valve (= pedicle valve): The valve in a brachiopod shell that contains the pedicle foramen.

Pedicle: A fleshy stalk which attaches most brachiopods (Brachiopoda) to the sea floor. It emerges either posteriorly from between the two valves of the shell or from a triangular notch (the delthyrium) in one of them. In some species it issues from a circular pedicle foramen (in the ventral valve). In other species it is either not developed or atrophied.

Dorsal valve (= brachial valve): The valve in a brachiopod shell that does not contain the pedicle foramen

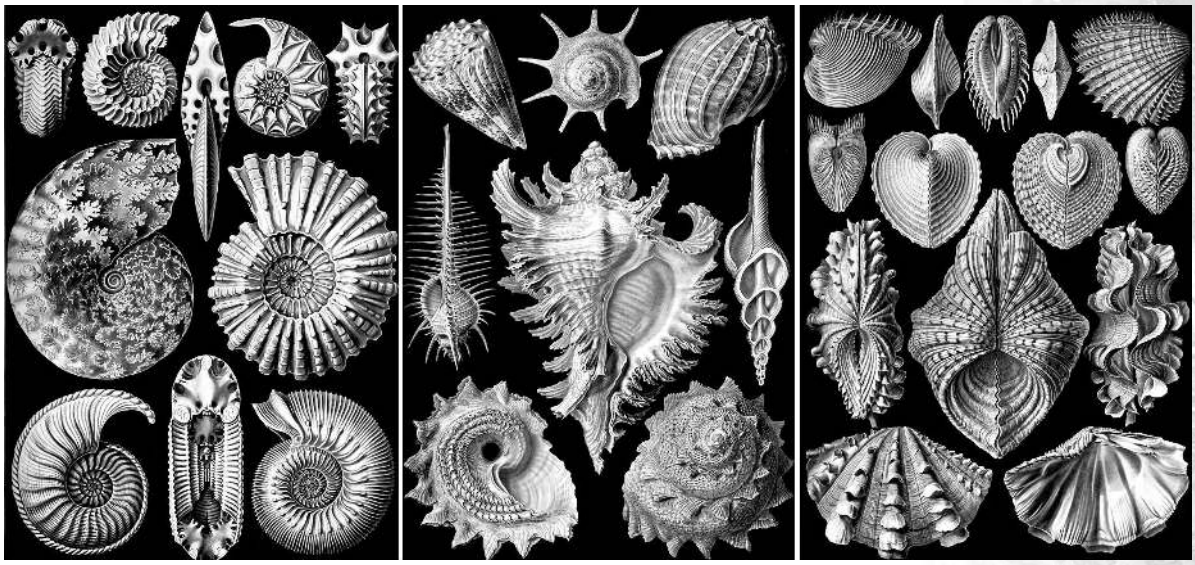




Lophophore: An organ characteristic of aquatic invertebrates of the phyla Bryozoa, Phoronida, and Brachiopoda that functions in filter feeding. It consists of a ridge of hollow tentacles bearing cilia, which waft food particles into the mouth.

Source: Carlson, S.J., 2016. The evolution of Brachiopoda. Annual Review of Earth and Planetary Sciences, 44, pp.409-438.

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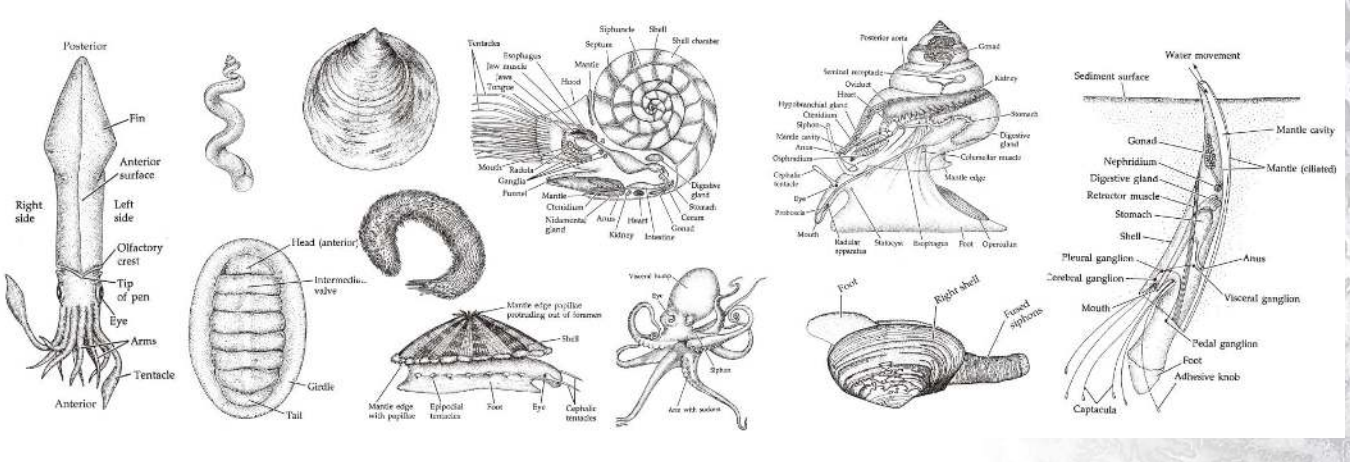


Macrofossils: Molluscs

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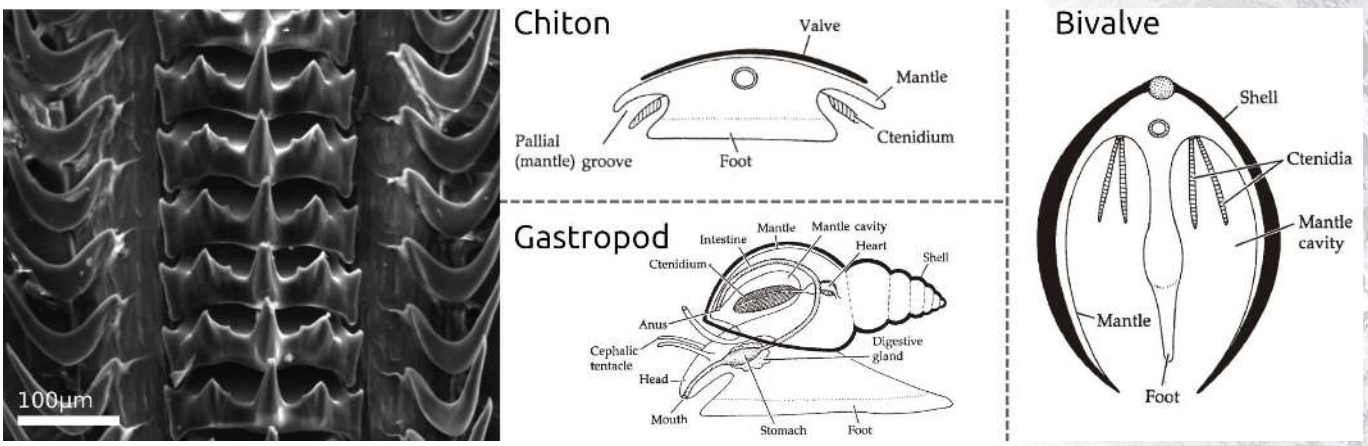
Introduction and biology

What is a Mollusc?



Source for all images: Brusca, R.C. and Brusca, G.J., 2003. Invertebrates Sinnauer Associates.

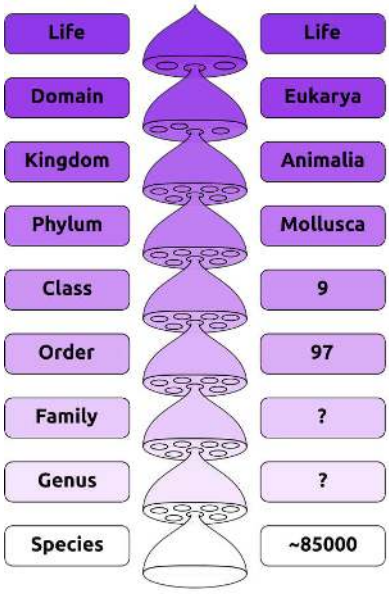
What makes a molluscs?

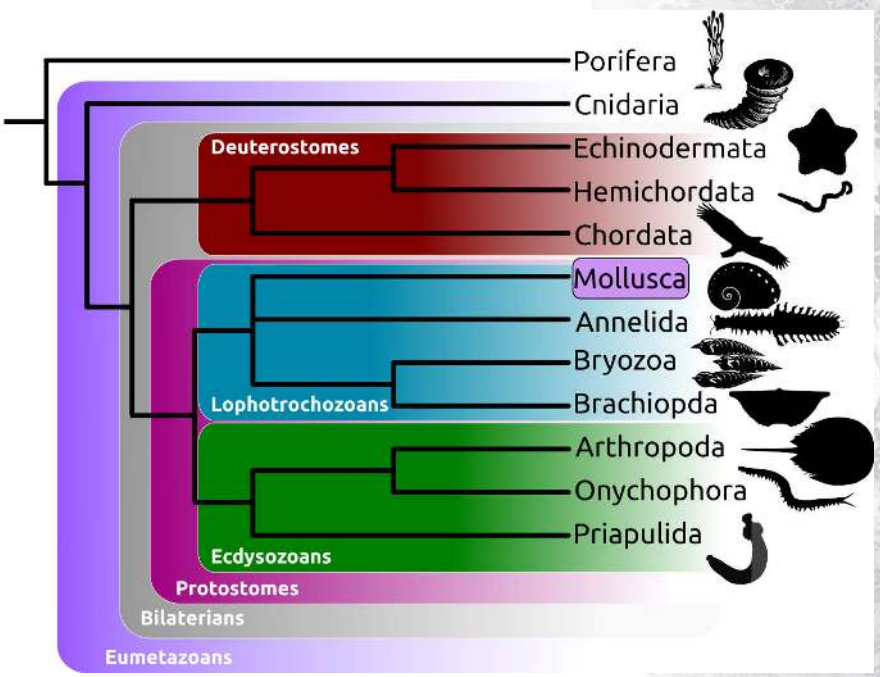


Mantle: The fold of skin covering the dorsal surface of the body of molluscs, which extends into lateral flaps that protect the gills in the mantle cavity (the space between the body and mantle). The outer surface of the mantle secretes the shell (in species that have shells).

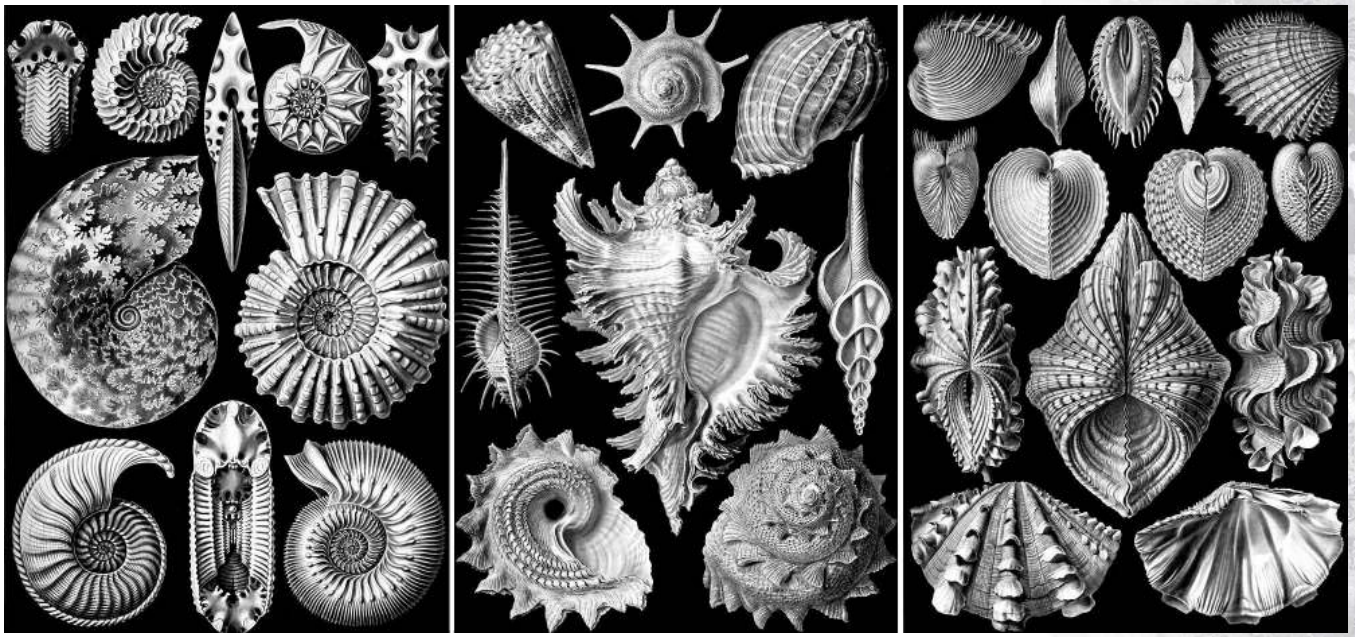
Gastropod radula: Fernando Archuby; Right: Brusca, R.C. and Brusca, G.J., 2003. Invertebrates Sinnauer Associates.

Where do they sit on the tree of life?



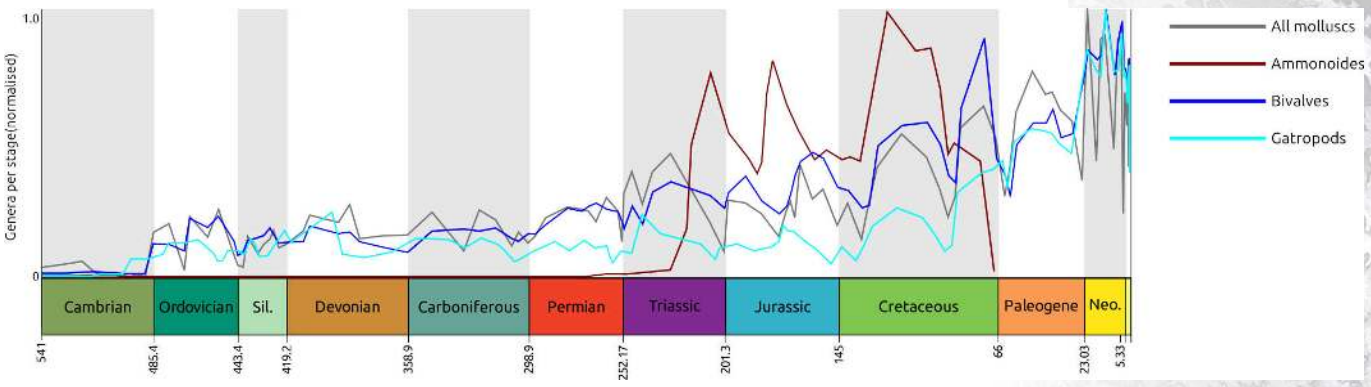


What molluscs are common in the fossil record?



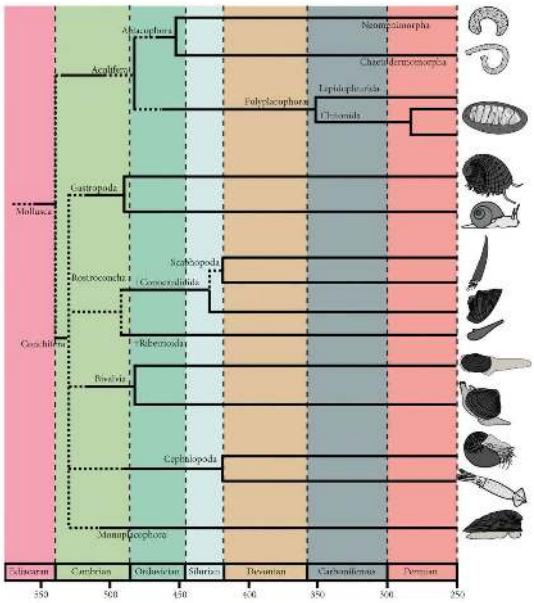
Cephalopods | Gastropods | Bivalves

When were the different groups of molluscs around?



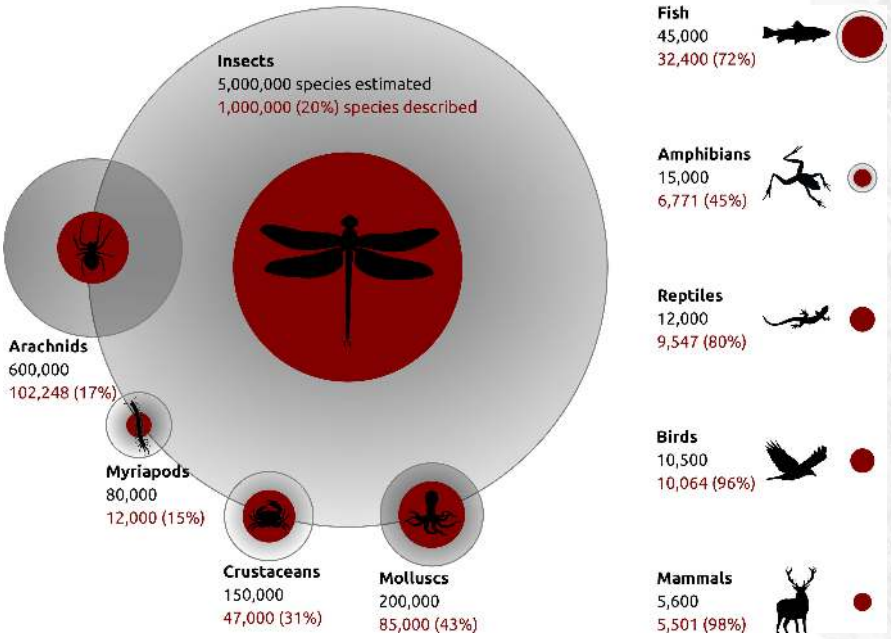
Fossil data from PBDB.

How are all these groups related?



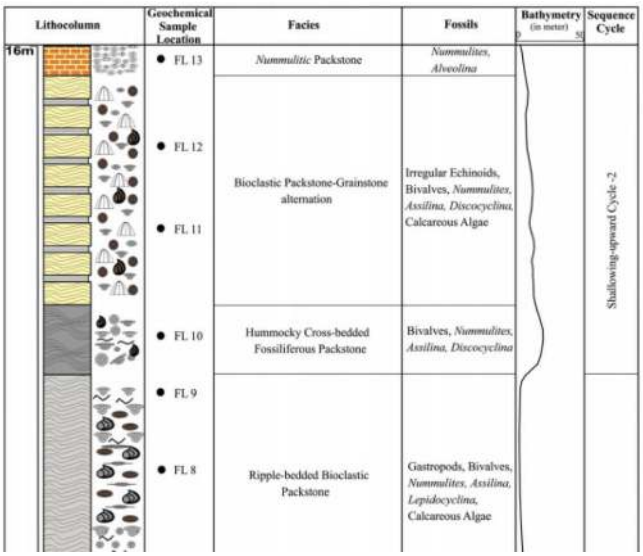
Source: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.

Why should I care?



Data from: Chapman, A.D., 2009. Numbers of living species in Australia and the world.

Why should I care?

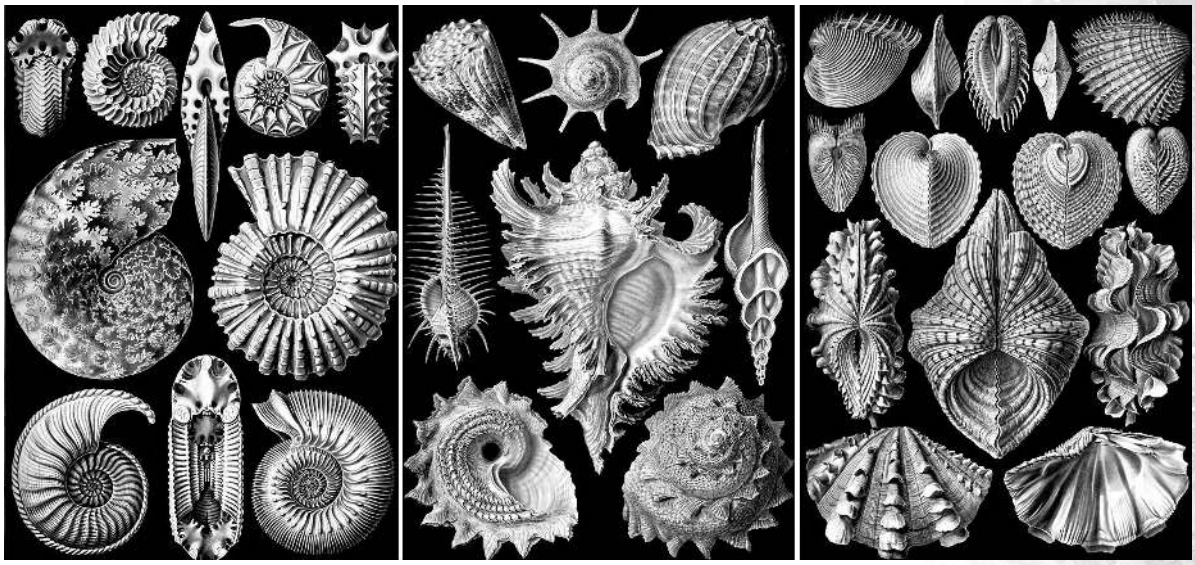




Left: Depositional environments and sources for the middle Eocene Fulra Limestone Formation, Kachchh Basin, western India: Evidences from facies analysis, mineralogy, and geochemistry

Right: CC0, Matmor Formation (Middle Jurassic), southern Israel.

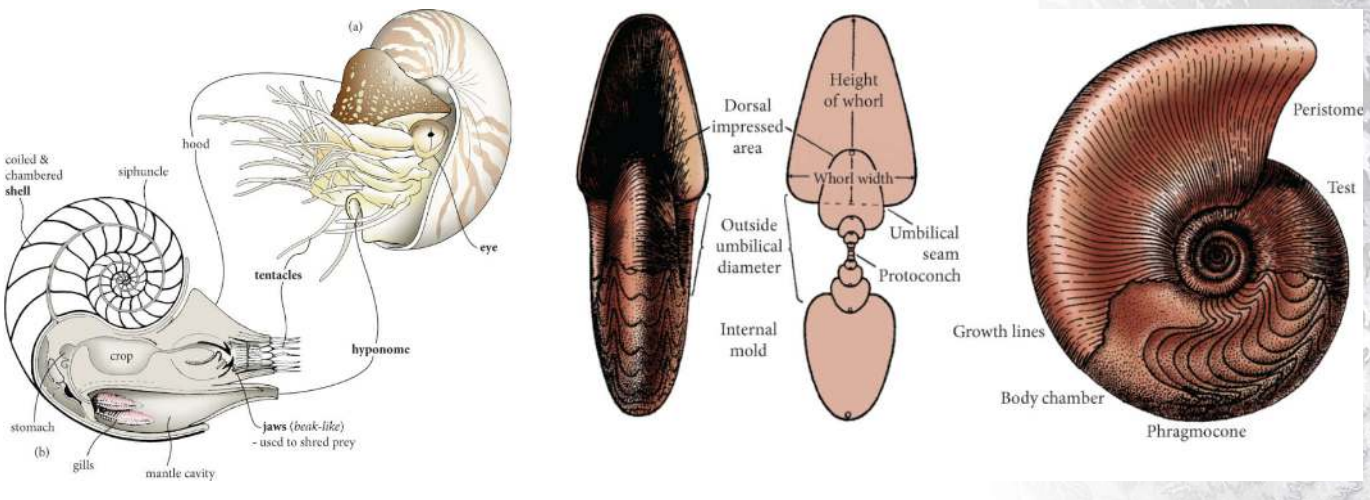
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Macrofossils: Molluscs

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Morphology / Anatomy
Cephalopoda



Left: Nautiloidea; Right: Ammonoidea.
Source: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons..

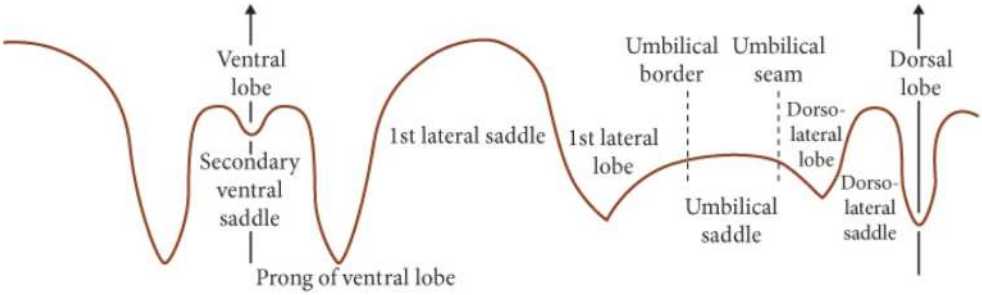
Phragmocone: The septate shell of a cephalopod (Cephalopoda). The term is also applied to the septate portion of a belemnite (Belemnitida) skeleton which, although reduced in size, is homologous to the external shell of other cephalopods.

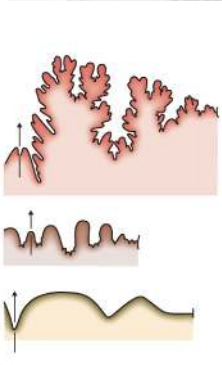
Siphuncle: A long tube present in those cephalopods (Cephalopoda) that possess external shells. It runs internally, through all the chambers, and contains the siphuncular cord of body tissue, extending from the visceral mass through a perforation in each septum of the shell. The siphuncle releases gas into the unoccupied chambers of the shell, making the animal buoyant.

Septa: The Cephalopoda shell is divided internally into a series of chambers separated by septa

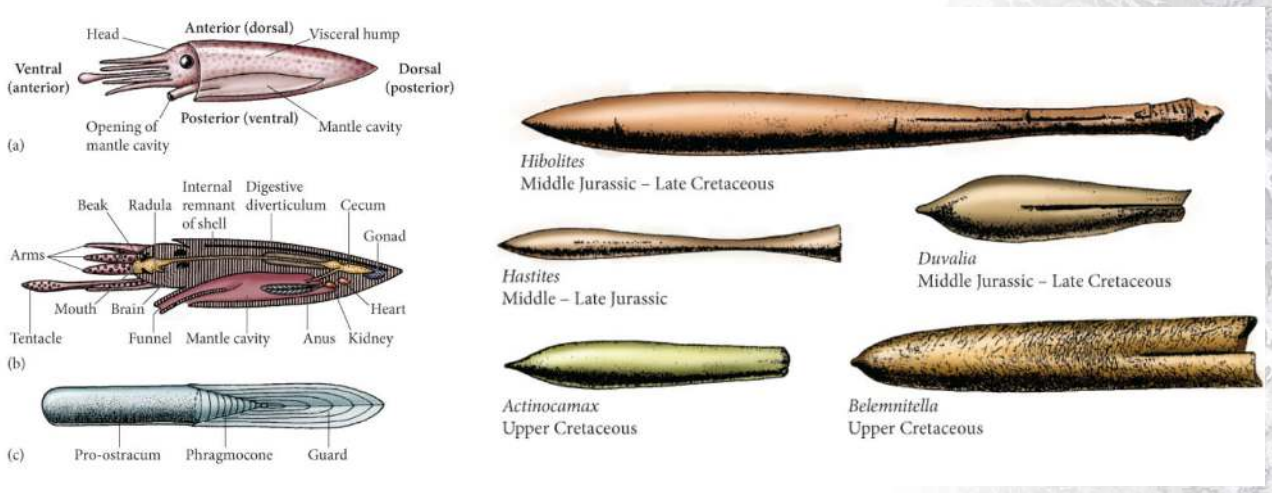
Suture line: Line formed where septa intersect with outer shell

Whorl: One of the coils in a molluscan shell. Whorls are most noticeable among Gastropoda and Cephalopoda, in which the shell is a hollow cone that grows only at the apertural end and tends to curl about its vertical axis. In most molluscs the coils form a helix in which each whorl is attached to the next, but in some the coiling is loose.





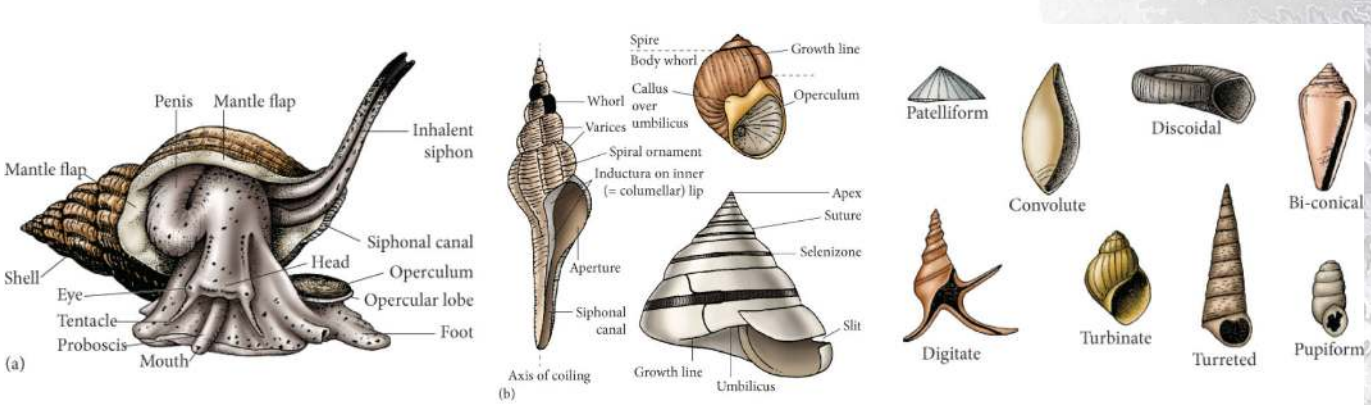
Source: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.



Belemnite (class Cephalopoda, subclass Coleoidea): One order of extinct cephalopods in which the shell is internal. Belemnites appear in the Jurassic, continue through the Cretaceous, and a few persist into the Eocene.

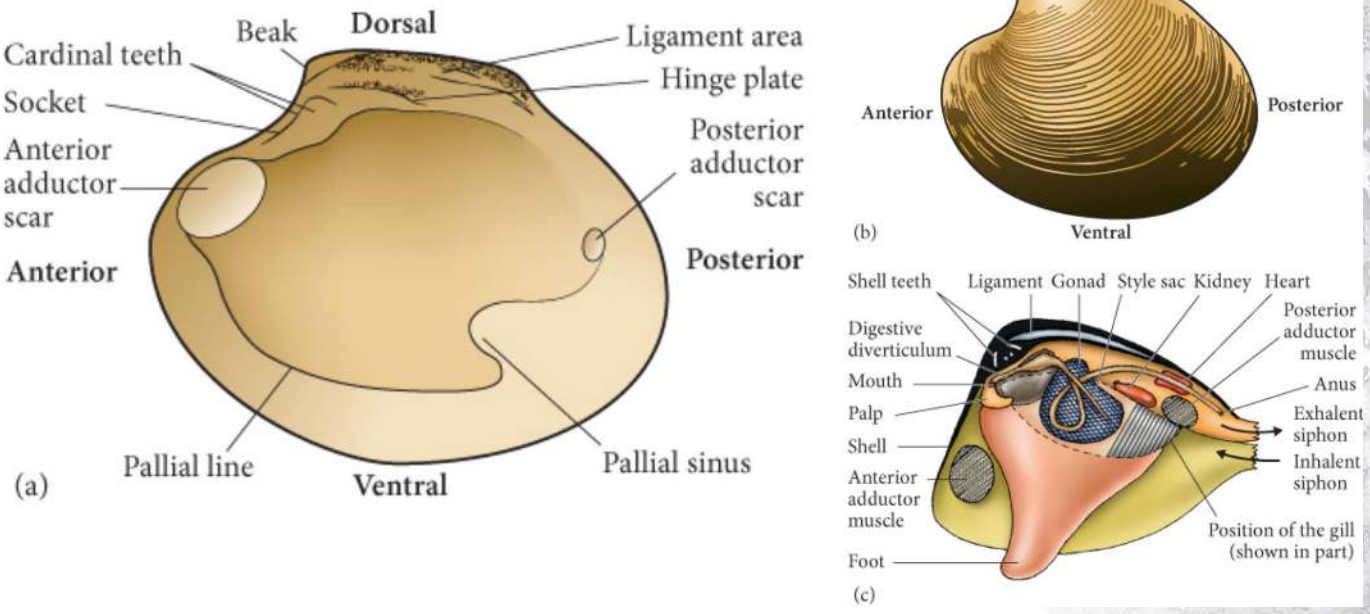
Source: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons.

Morphology / Anatomy
Gastropoda



Aperture: In a mollusc (Mollusca) shell, the opening through which the soft parts of the animal emerge.
Torsion: In Gastropoda, the twisting of the body through 180 degrees so that the digestive and nervous systems have a U shape and the mantle cavity, anus, gills, and two nephridiopores occupy an anterior position behind the head.
Source: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons..

Morphology / Anatomy
Bivalvia



Source: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons..

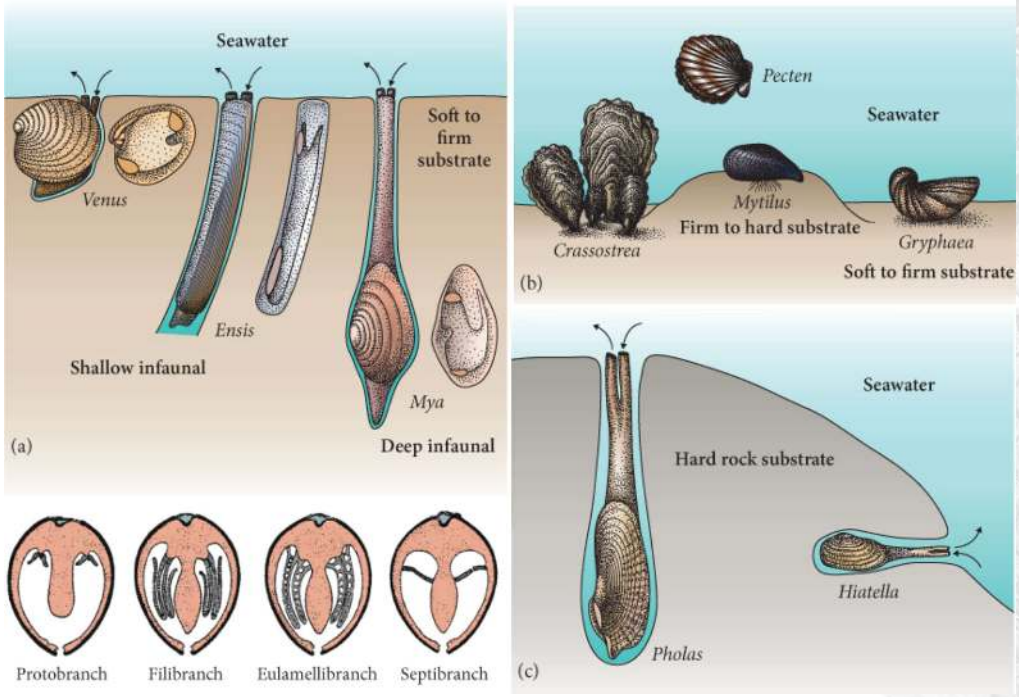
Commissure: A line or plane of junction, e.g. between the two valves of the shell in a brachiopod (Brachiopoda) or bivalve (Bivalvia).

Pallial line: Connecting the two scars on the interior of the valve there is the pallial line, a linear depression marking the inner margin of the thickened mantle edges that may be marked by an inward deflection at the posterior part, the pallial sinus.

Pallial sinus: In certain bivalves (Bivalvia), a recess or embayment in the pallial line; most bivalves with a pallial sinus are burrowers.

Umbo: A convex protuberance or swelling in the centre. In Brachiopoda and Bivalvia, the first part of the shell to be formed.

Adductor scar: A depressed or raised marking on the interior of a shell, usually of a brachiopod (Brachiopoda) or bivalve (Bivalvia), which was the point of attachment of a muscle.



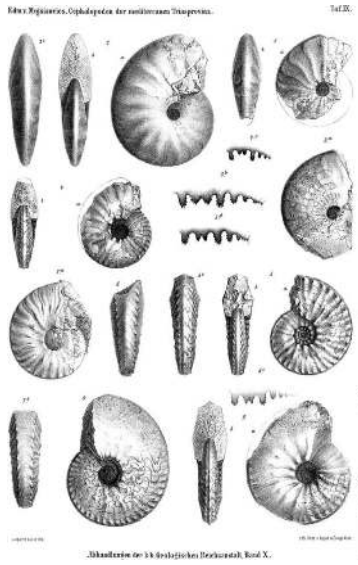
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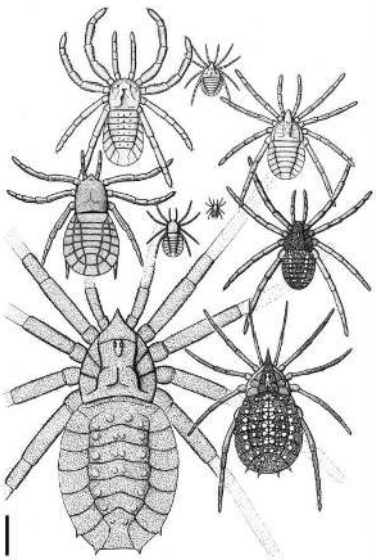


Rudist bivalve: An extinct group of coral- and horn-shaped bivalves (Bivalvia, subclass Heterodonta). They have a variable morphology and many species are not easily seen to be bivalves. They are adapted to a sedentary mode of life. Some forms have a larger lower left valve and an upper right valve that forms a flat lid; other forms have a larger lower right valve.

Source: Benton, M.J. and Harper, D.A., 2020. Introduction to paleobiology and the fossil record. John Wiley & Sons | Allaby, M., 2013. A dictionary of geology and earth sciences. Oxford University Press.

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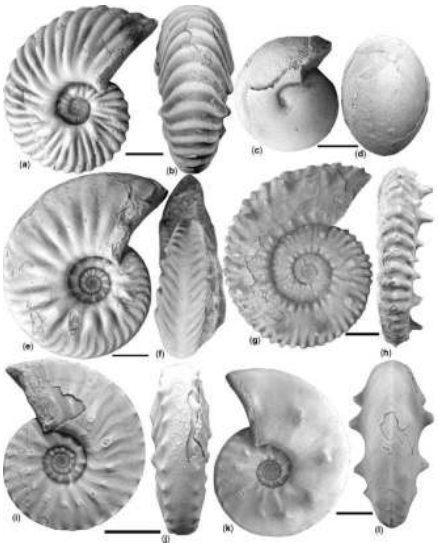


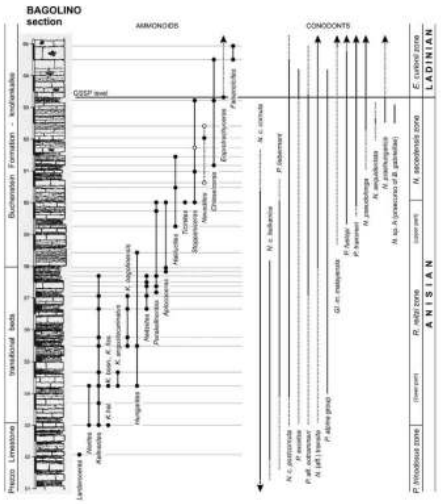


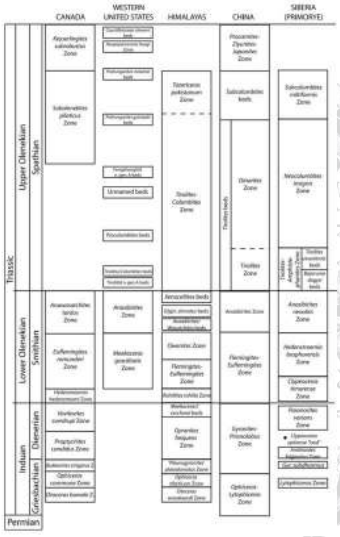
Macrofossils: Biostratigraphy – 1

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What is biostratigraphy?



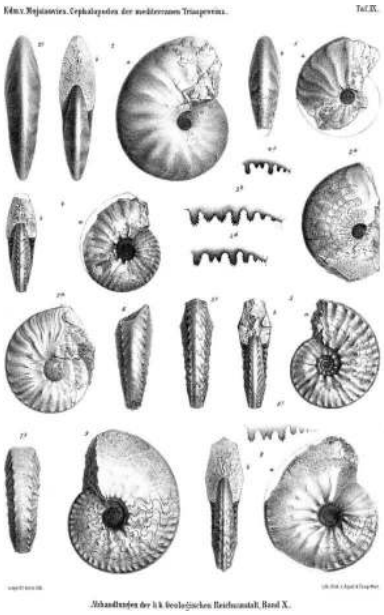


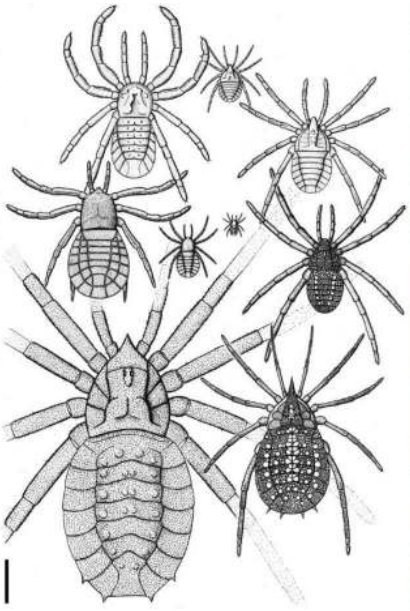


Biostratigraphy: Branch of stratigraphy that involves the use of fossil organisms in the dating and correlation of the stratigraphic sequences of rock in which they are discovered. A zone is the fundamental division recognized by biostratigraphers.

Modified from: Allaby, M., 2013. A dictionary of geology and earth sciences. Oxford University Press.
Image source: Balini, M., Lucas, S.G., Jenks, J.F. and Spielmann, J.A., 2010. Triassic ammonoid biostratigraphy: an overview. Geological Society, London, Special Publications, 334(1), pp.221-262.

What are the assumptions underlying biostratigraphy?

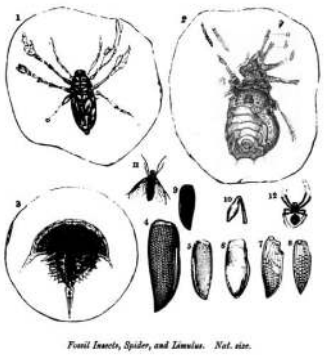






Lazarus taxon: A taxon that disappears from the fossil record close to an extinction horizon, but reappears again much later in the sequence.
Source: Allaby, M., 2013. A dictionary of geology and earth sciences. Oxford University Press.

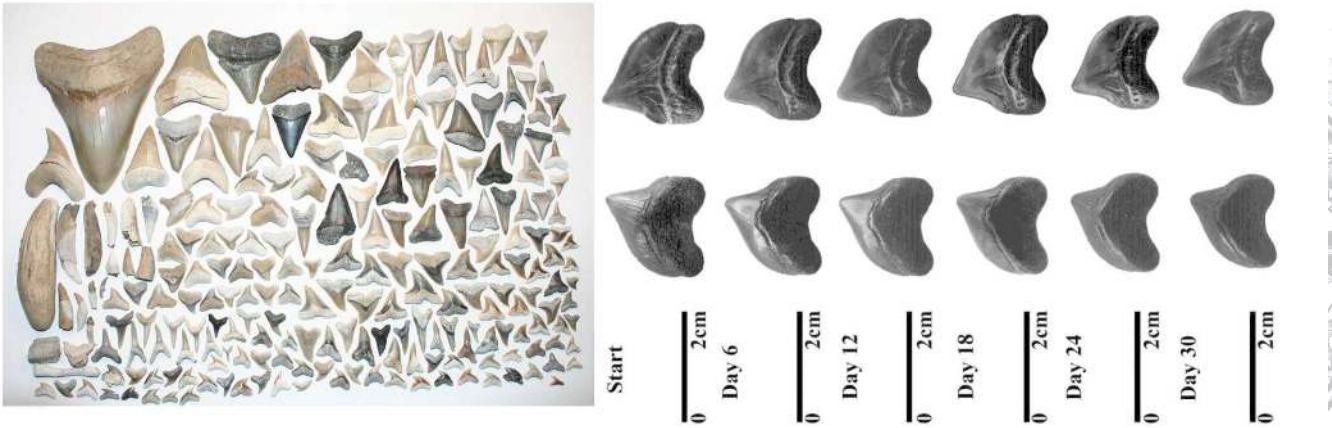




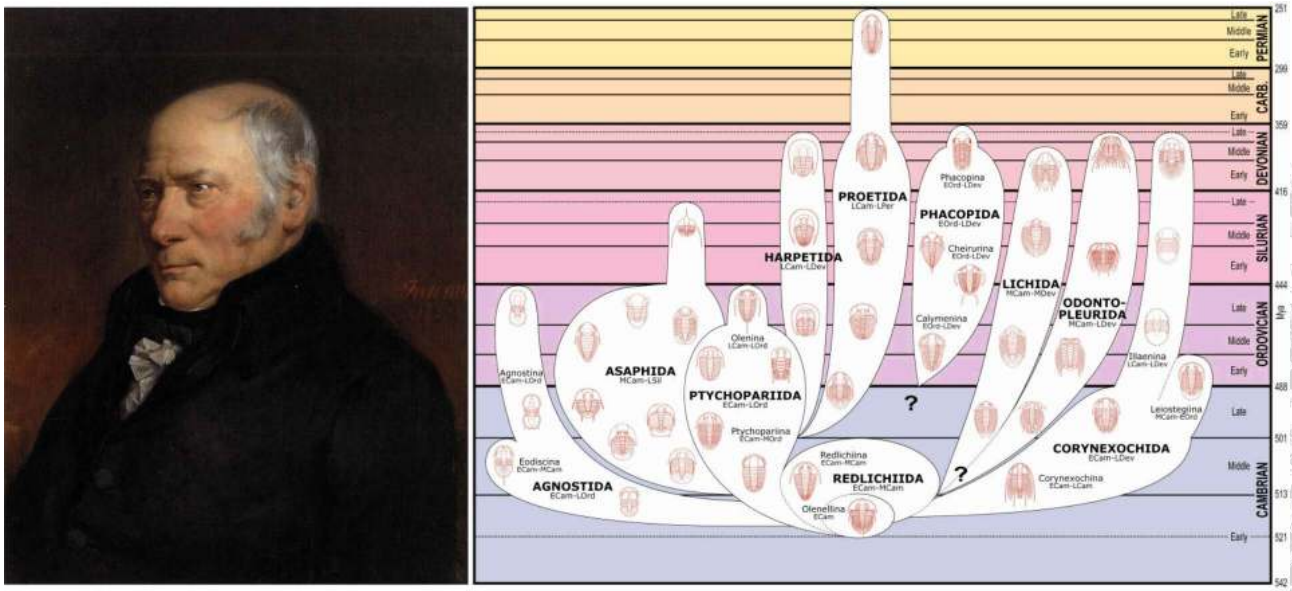
Fossil Insects, Spider, and Limulus. Nat. size.



Ricinulei image courtesy of Marshal Hedin.



Source: Becker, M.A. and Chamberlain, J.A., 2012. Squalicorax chips a tooth: a consequence of feeding-related behavior from the lowermost Navesink Formation (Late Cretaceous: Campanian-Maastrichtian) of Monmouth County, New Jersey, USA. Geosciences, 2(2), pp.109-129.



Faunal succession: The principle, first recognized at the beginning of the 19th century by William Smith, that different strata each contain particular assemblages of fossils by which the rocks may be identified and correlated over long distances; and that these fossil forms succeed one another in a definite and habitual order.

Source: Allaby, M., 2013. A dictionary of geology and earth sciences. Oxford University Press.

So, what is our fundamental unit?

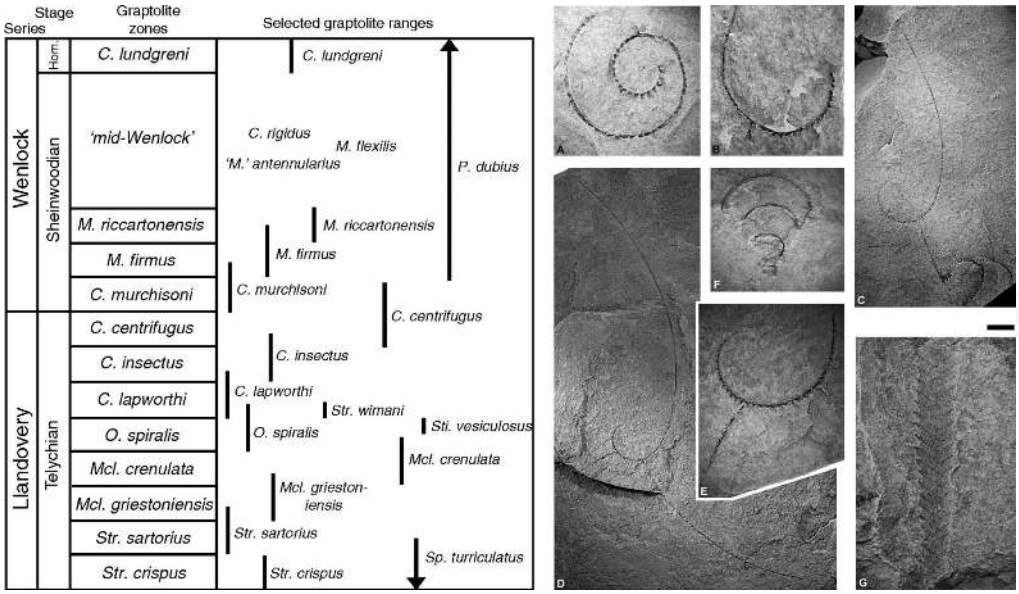
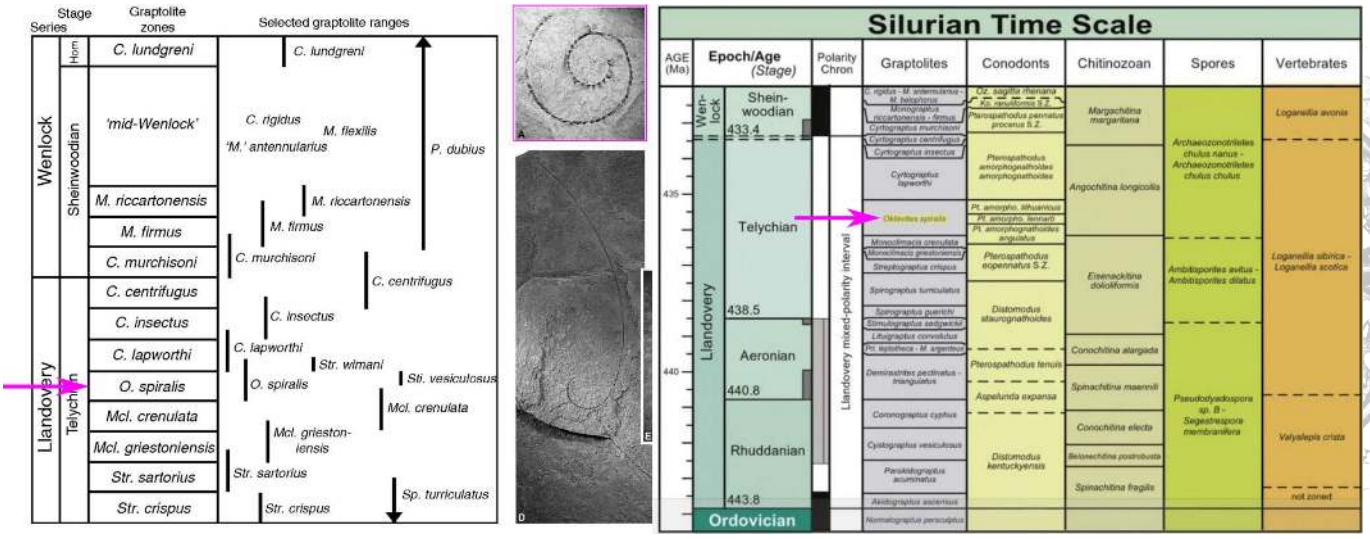


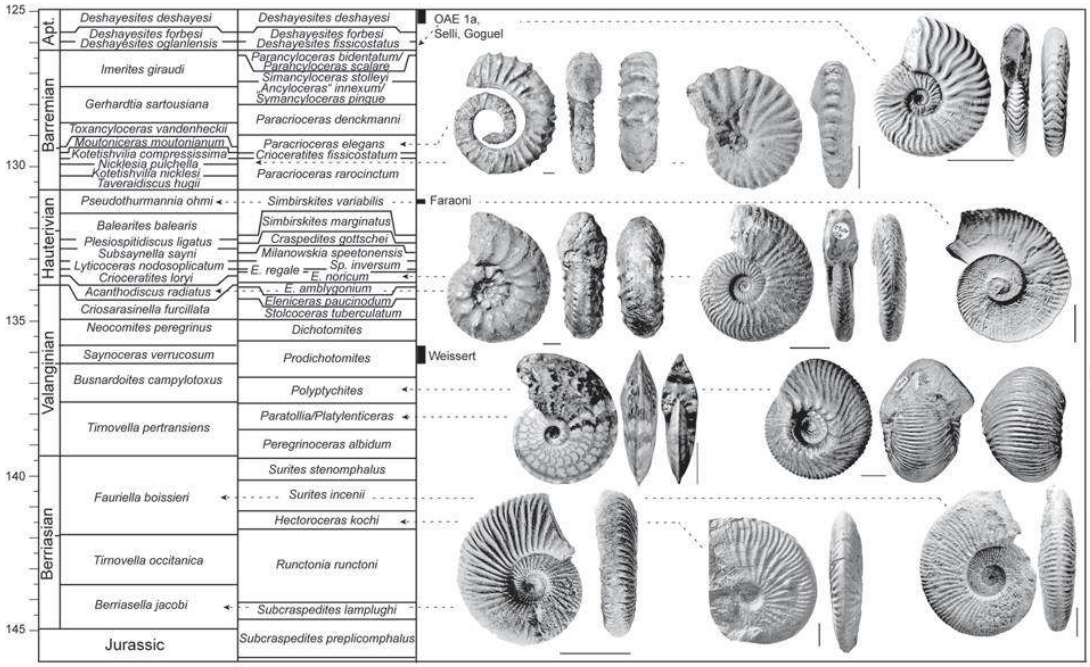
Image source: Cramer, B.D., et al. 2010. Testing the limits of Paleozoic chronostratigraphic correlation via high-resolution (< 500 ky) integrated conodont, graptolite, and carbon isotope (δ13Ccarb) biochemostratigraphy across the Llandovery–Wenlock (Silurian) boundary: Is a unified Phanerozoic time scale achievable? Geological Society of America Bulletin, 122(9-10), pp.1700-1716.

Arvestål, E.H. and Streng, M., 2013. Cyrtograptids from the Telychian (upper Llandovery) of Kinnekulle Mountain, southern Sweden. GFF, 135(1), pp.45-56.

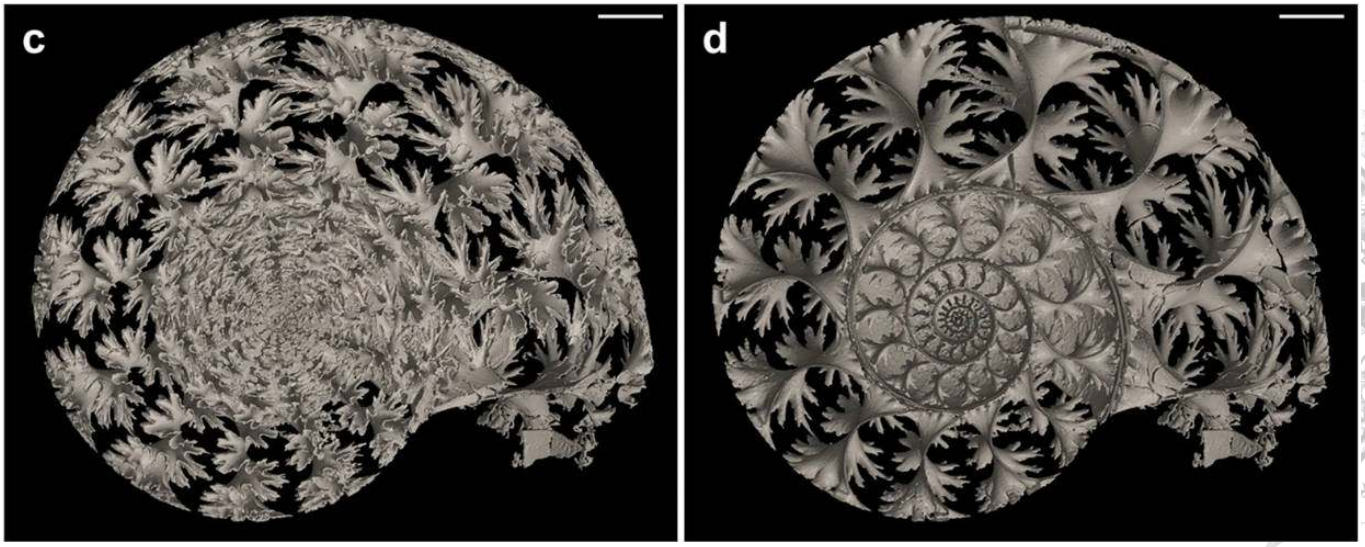
What is an index fossil?



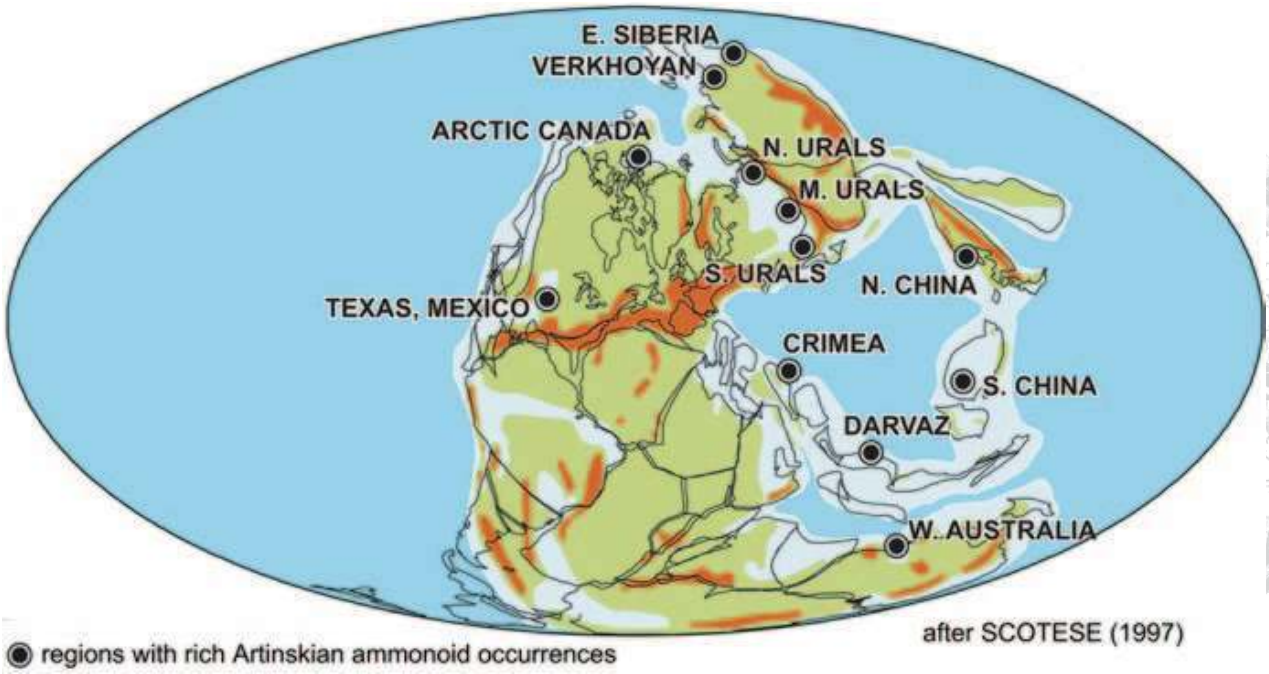
Index fossil: Fossil whose presence is chosen to denote the zone in which it occurs and after which the zone is named.



Index fossil: Fossil whose presence is chosen to denote the zone in which it occurs and after which the zone is named.
Lehmann, J., 2015. Ammonite biostratigraphy of the Cretaceous—an overview. In Ammonoid paleobiology: from macroevolution to paleogeography (pp. 403-429). Springer, Dordrecht.



Source: Inoue, S. and Kondo, S., 2016. Suture pattern formation in ammonites and the unknown rear mantle structure. Scientific reports, 6, p.33689.



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Dieter, K. & DaBeats, K. 2015. Biogeography of Paleozoic ammonoids. In Ammonoid Paleobiology: From macroevolution to paleogeography (pp. 145-161). Springer, Dordrecht.



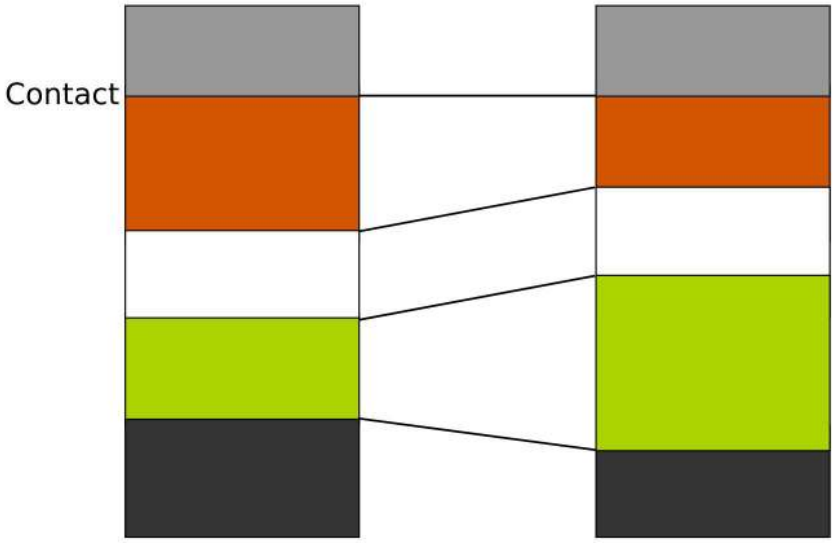
Photo by Lee R. Berger.

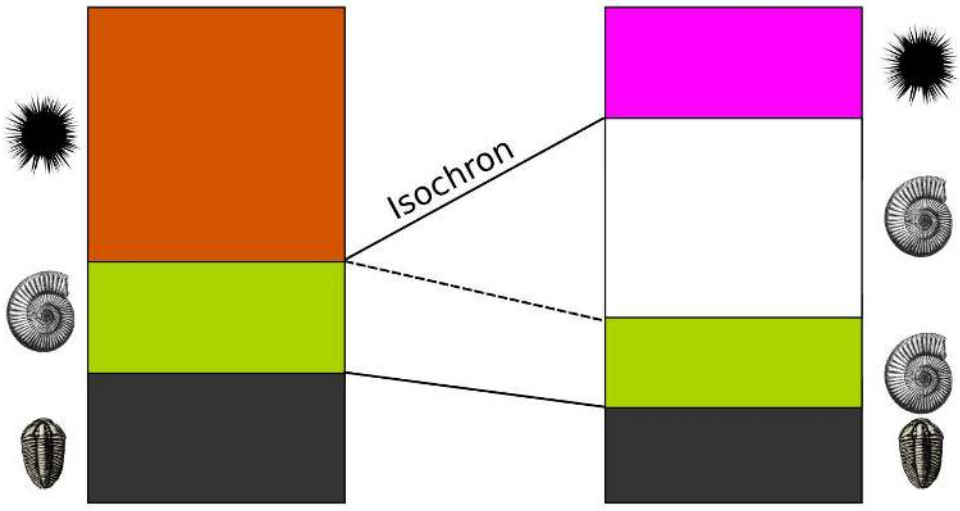


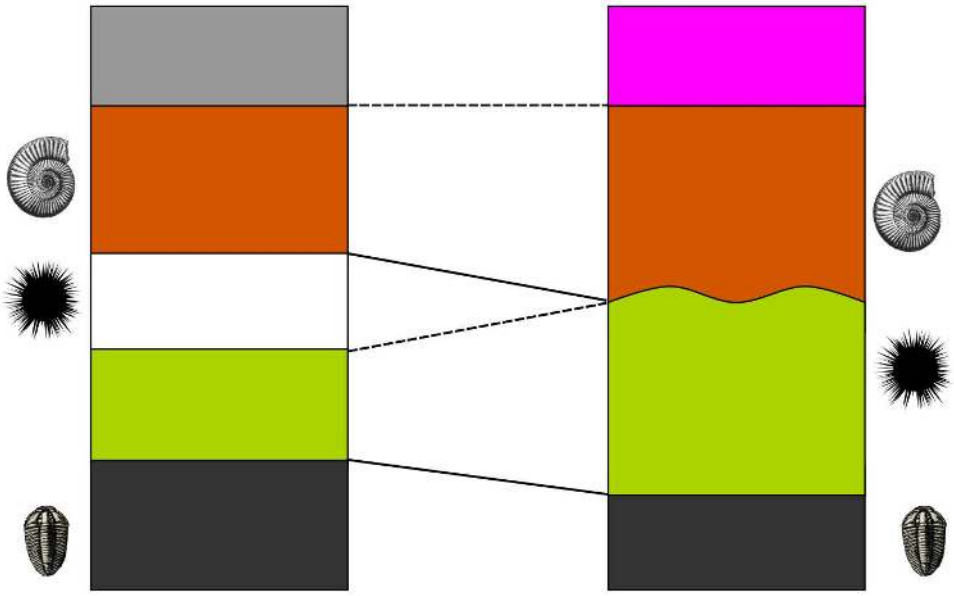
Index fossil: Fossil whose presence is chosen to denote the zone in which it occurs and after which the zone is named..



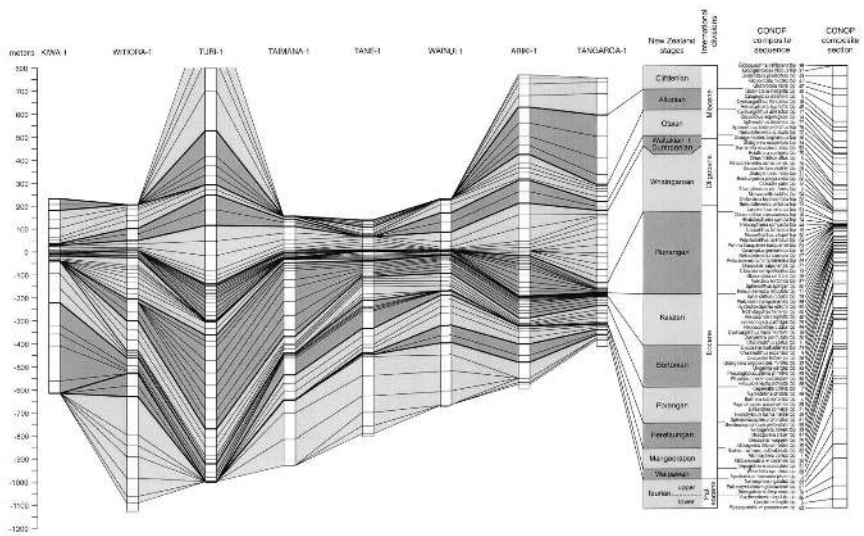
How can can we use fossils to correlate rocks?







Biostratigraphy - Sections



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